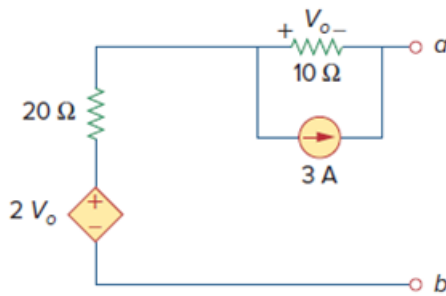


Problem

Find the Norton equivalent at terminals a - b of the circuit in Fig. 4.119.

Figure 4.119



Step-by-step solution

Step 1 of 7

Refer to circuit diagram in Figure 4.119 in the text book.

Consider the circuit diagram shown in Figure 1 to calculate the Thevenin's voltage.

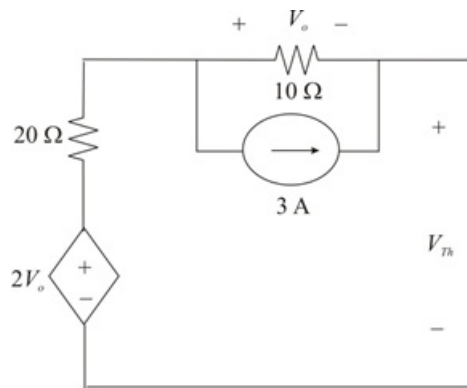


Figure 1

Step 2 of 7

Notice that no current passes through the $20\ \Omega$ resistor since the circuit is opened.

Calculate the value of voltage V_o .

$$\begin{aligned} V_o &= (-3)(10) \\ &= -30\ \text{V} \end{aligned}$$

Consider the expression for the Thevenin's Voltage V_{Th} .

$$\begin{aligned} V_{Th} &= -V_o + 2V_o \\ &= V_o \end{aligned}$$

Substitute $-30\ \text{V}$ for V_o .

$$V_{Th} = -30\ \text{V}$$

Therefore, the value of the Thevenin's voltage V_{Th} is $\boxed{-30\ \text{V}}$.

Step 3 of 7

Short the output terminals to calculate the short circuit current.

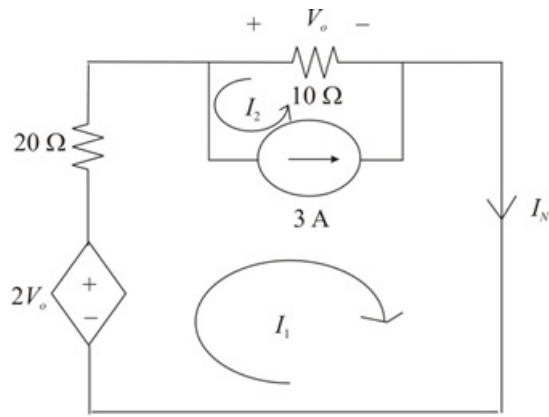


Figure 2

Step 4 of 7

Calculate the value of voltage V_o .

$$V_o = -I_2(10)$$

Therefore, the value of the voltage V_o is, $-I_2(10)$.

Consider the expression for the current I_2 .

$$I_2 = 3 - I_1$$

Therefore, the expression for the current I_2 is, $3 - I_1$.

Step 5 of 7

Apply super mesh analysis in loops 1 and 2.

$$-2V_o + I_1(20) - I_2(10) = 0$$

Substitute $-I_2(10)$ for V_o .

$$(-2)(-I_2)(10) + I_1(20) - I_2(10) = 0$$

$$I_2(20) + I_1(20) - I_2(10) = 0$$

$$I_2(10) + I_1(20) = 0$$

Substitute $3 - I_1$ for I_2 .

$$(3 - I_1)(10) + I_1(20) = 0$$

$$30 - 10I_1 + 20I_1 = 0$$

$$30 + 10I_1 = 0$$

$$10I_1 = -30$$

$$I_1 = -\frac{30}{10}$$

$$= -3 \text{ A}$$

Here, I_1 represents the short circuit current I_N .

$$I_N = -3 \text{ A}$$

Therefore, the value of the short circuit current I_N is -3 A .

Step 6 of 7

Calculate the value of the value of the Norton's resistance.

$$R_N = \frac{V_{Th}}{I_N}$$

Substitute -3 A for I_N and -30 V for V_{Th} .

$$\begin{aligned} R_N &= \frac{-30}{-3} \\ &= 10 \, \Omega \end{aligned}$$

Therefore, the value of the Norton's resistance is $10 \, \Omega$.

Step 7 of 7

Draw the Norton's equivalent circuit diagram.

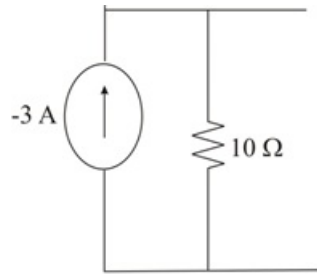


Figure 3

Therefore, the Norton's circuit is shown in Figure 3.